ENTHUSE, LEADER & ACHIEVER COURSE

PHASE : ALL SRG ENTHUSE, LEADER, ACHIEVER COURSE & LAKSHYA PHASE

TARGET : PRE-MEDICAL : 2026

Test Type : **SRG-MAJOR** Test Pattern : **NEET (UG)**

TEST DATE : 15-06-2026

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	2	2	4	2	4	1	3	2	2	1	3	3	3	1	3	2	2	3	3	4	4	3	4	1	2	1	2	2	3	3
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	3	1	2	4	2	1	4	1	3	2	3	4	1	2	3	1	4	1	4	1	1	1	4	3	1	3	3	3	1	4
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	3	3	4	4	1	2	2	2	2	2	4	3	1	1	2	3	3	2	2	2	3	3	4	4	4	2	1	4	3	4
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	4	4	1	3	2	3	4	3	3	1	2	3	4	1	1	3	4	4	4	3	2	4	4	3	1	1	1	3	4	1
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	1	3	3	2	3	3	3	2	2	1	2	1	3	1	3	4	2	4	1	2	4	3	2	2	2	3	4	3	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	1	1	4	1	4	4	4	3	2	3	4	2	3	1	3	1	1	3	3	2	1	3	3	1	1	1	1	2	4	4

HINT – SHEET

1. Ans (2)

$\vec{r} = -\sqrt{2}\left(\cos t\hat{i} + \cos t\hat{j}\right)$

$\vec{v} = \frac{d\vec{r}}{dt} = \sqrt{2}\left(\sin t\hat{i} + \sin t\hat{j}\right)$

$\vec{a} = \sqrt{2}\left(\cos t\hat{i} + \cos t\hat{j}\right)$

$\vec{F} = m\vec{a} = 10\sqrt{2}\left(\cos t\hat{i} + \cos t\hat{j}\right)$

Angle between velocity & acceleration is

$\cos\theta = \frac{2(2\sin t\cos t)}{4\sin t\cos t} = 1$

$\Rightarrow \cos\theta = 1$

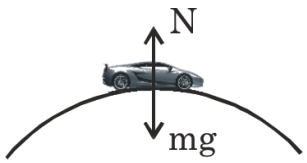
$\Rightarrow \theta = 0°$

So path is straight line

2. Ans (2)

F_{net} towards centre $= mv^2/R$

At A



$mg - N = \frac{mv^2}{R_A}$

$N > 0 \Rightarrow g > v^2/R_A$

At B $N - mg = \frac{mv^2}{R_B}$

$N_B = mg + \frac{mv^2}{R_B}$

It is not necessary that $\frac{v^2}{R_B}$ should be less than g.

3. Ans (4)

$E(t) = A^2 e^{-\alpha t}$

$\ln E = \ln A^2 + \ln e^{-\alpha t}$

$\ln E = 2 \ln A - \alpha t$

$\frac{\Delta E}{E} \times 100 = 2\frac{\Delta A}{A} \times 100 + \alpha \Delta t \times 100$

$= 2 \times 1.25 + \frac{2}{10} \times 1.5 \times 5$

$= 2.5 + 1.5 = 4\%$

4. Ans (2)

Mean diameter $\bar{d} = \frac{d_1 + d_2 + d_3}{3}$.

$\bar{d} = \frac{1.002 + 1.004 + 1.006}{3} = \frac{3.012}{3} = 1.004$ cm

Absolute error in each reading :

$\Delta d_1 = |d_1 - \bar{d}| = |1.002 - 1.004| = 0.002$ cm

$\Delta d_2 = |d_2 - \bar{d}| = |1.004 - 1.004| = 0$ cm

$\Delta d_3 = |d_3 - \bar{d}| = |1.006 - 1.004| = 0.002$ cm

Mean absolute error in diameter ($\overline{\Delta d}$) :

$\overline{\Delta d} = \frac{1.002 + 0 + 0.002}{3} = \frac{1.004}{3} = 0.001$ cm

Mean absolute error in radius (Δd)

$\overline{\Delta d} = \frac{0.001}{2} = 0.0005$ cm

Percentage error in the measurement of diameter :

$\%$ error $= \frac{\text{Mean absolute error}}{\text{Mean diameter}} \times 100$

$= \frac{0.0005}{1.004} \times 100 = 0.049\%$

5. Ans (4)

At t_1, tangent having $\theta < 90°$

$\therefore$ a = +ve & v = +ve

$\therefore \vec{a} \cdot \vec{v}$ is positive

At t_2, a = 0 & v = +ve $\Rightarrow \vec{a} \cdot \vec{v} = 0$

At t_3, a = −ve & v + ve $\Rightarrow \vec{a} \cdot \vec{v} =$ −ve

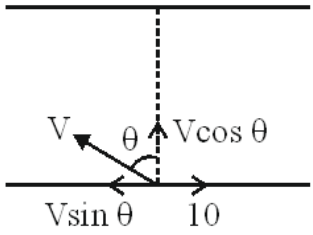
If angle between $\vec{v}$ & $\vec{a}$ is less than 90° then particle speeds up.

6. Ans (1)

No. of taxi = $\frac{240}{10}$ = 24 but when 24th starts motion when that taxi reach the destination so it will meet 23 only.

7. Ans (3)

Let speed of ant relative to belt is V, if ant reaches at B

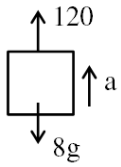


$V \sin\theta = 10$

$V \cos\theta = \frac{50}{10} = 5$

$V = \sqrt{10^2 + 5^2} = 5\sqrt{5}$ cm/s

8. Ans (2)



FBD of 8 kg block

$a = \frac{120 - 80}{8} = 5$ m/s^2

9. Ans (2)

Work doesn't depend on path for conservation force

$W_{AC} = 2J$, So $W_{CA} = -2J$

$W_{AB} + W_{BE} + W_{EA} = 0$

$1 + (-3) + W_{EA} = 0$

$W_{EA} = 2$

$W_{AE} = -2J$

$W_{DF} + W_{FB} + W_{BC} = W_{DC}$

$-1 + 2 + 1 = W_{DC}$

$W_{DC} = 2J$

$W_{BF} + W_{FD} = W_{BD}$

$-2 + 1 = W_{BD}$

$W_{BD} = -1J$

10. Ans (1)

$U = 40 + 20 \sin 4\pi x$

At $x = \frac{9}{8}m = U = 40 + 20\sin\left[4\pi \times \frac{9}{8}\right]$

$= 40 + 20\sin\left[\frac{9\pi}{2}\right]$

$= 60$ J

At $x = \frac{5}{8}m$

$U = 40 + 20 \sin\left[4\pi \times \frac{5}{8}\right]$

$U = 40 + 20 \sin\left[\frac{5\pi}{2}\right]$

$U = 40 + 20 = 60$ J

At both $x = \frac{9}{8}m$ and $x = \frac{5}{8}m$, U = 60 J Now

If total ME = 60 J, then KE = 0 [minimum]

Thus, KE is minimum at $x = \frac{5}{8}m$ & $\frac{9}{8}m$ ....(C)

Now $F = \frac{-dU}{dx}$

$= \frac{-d}{dx}[40 + 20\sin 4\pi x]$

$F = -80\ \pi \cos[4\pi x]$

At $x = \frac{9}{8}m$

$F = -80\ \pi \cos\left[4\pi \times \frac{9}{8}\right]$

$F = -80\ \pi \cos\left[\frac{9\pi}{2}\right]$

F = 0 Thus at $x = \frac{9}{8}m$ particle is in equilibrium

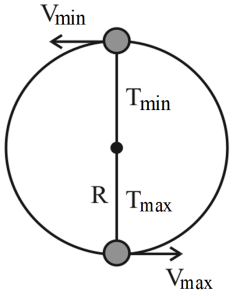
Options (A) & (C) are correct

11. Ans (3)

$T_{max} - mg = \frac{mv_{max}^2}{R}$

$mg + T_{min} = \frac{mv_{min}^2}{R}$

$\frac{T_{max}}{T_{min}} = \frac{\frac{v_{max}^2}{R} + g}{\frac{v_{min}^2}{R} - g} = \frac{\frac{v_{min}^2 + 2.g.2R}{R} + g}{\frac{v_{min}^2}{R} - g}$



$\Rightarrow v_{min} = 10$ m/s.

12. Ans (3)

$P_{max} = V \times I_{max} = 0.27$

$9 \times I_{max} = 0.27$

$I_{max} = 0.03$ A

$\therefore (R_S)_{min} = \frac{12 - 9}{0.03} = 100\ \Omega$

13. Ans (3)

Case–I :

$V_A < V_B$ $[\because (-10V) < (-5V)]$

D_1 & D_3 are RB and D_2 is FB

$\therefore R_1 = \frac{R}{4} + R + \frac{R}{4} = \frac{3R}{2}$

Case–II :

$V_A > V_B$

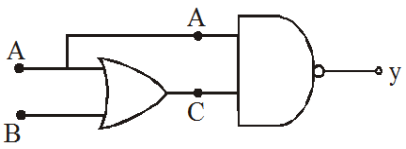
$\Rightarrow D_1$ & D_2 are FB and D_3 is RB

$\therefore R_2 = \frac{R}{4} + \frac{\frac{R}{2} \times \frac{R}{2}}{\left(\frac{R}{2} + \frac{R}{2}\right)} + \frac{R}{4} = \frac{3R}{4}$

$\Rightarrow$ same as case–II

Case–III : $V_A > V_B$ $R_3 = \frac{3R}{4}$

14. Ans (1)



C = A + B

and $y = \overline{A.C}$

A	B	C = (A + B)	A.C.	$y = \overline{A.C}$
0	0	0	0	1
0	1	1	0	1
1	0	1	1	0
1	1	1	1	0

15. Ans (3)

From given data

$E_{ph} = 2 + 5$

$E_{ph} = 7$ eV

and $10^{-5} = n_{ph} \times 10^{-5} \times 1.6 \times 10^{-19}$

$n_{ph} = 10^{20/16}$

$Power = W = \frac{n_{ph}}{t} \times E$

$W = \frac{10^{20}}{16} \times 7 \times 1.6 \times 10^{-19}$ Watt

W = 7 Watt

16. Ans (2)

Total energy of hydrogen atom in ground state = –13.6 eV

$\therefore M_H = M_P + M_e - \frac{13.6eV}{c^2}$

17. Ans (2)

$R \propto A^{1/3}$

$V = \frac{4}{3}\pi R^3 \propto A$

Mass $\propto$ A

So density is independent of A.

18. Ans (3)

$m_1v_1 - m_2v_2 = (m_1 + m_2)v_C$

$\frac{h}{\lambda_1} - \frac{h}{\lambda_2} = \frac{h}{\lambda'}$

$\lambda' = \frac{\lambda_1\lambda_2}{|\lambda_1 - \lambda_2|}$

19. Ans (3)

Photo-electric effect $\longrightarrow$ Particle nature of light

Davision-Germer Experiment $\longrightarrow$ Wave nature of electron

Franck-Hertz experiment $\longrightarrow$ Discrete energy levels of atom

20. Ans (4)

Density of water is maximum at 4°C

21. Ans (4)

$\lambda \propto \frac{1}{T}$

$\Rightarrow T_1 > T_2 > T_3$

Area under the curve $E\lambda$ & λ gives intensity so

$\sigma A e_3 T_3^4 > \sigma A e_2 T_2^4$

Now as $T_3 < T_2 \Rightarrow e_3 > e_2$

22. Ans (3)

On giving heat both sphere expand, sphere A expand in upward direction and B in downward, hence potential energy of A increases while potential energy of B decreases hence heat given to A not only increases temp but also PE, so temperature of A increase less than B.

23. Ans (4)

Number of collision at any time is equal for ABCD & EFGH, but change in momentum at ABCD is 2mv but at EFGH is mv.

This is because force EFGH absorbs totally all the molecules incident on it. These molecules do not rebound. Change in momentum causing force and hence pressure is reduced to half compared to force || pressure on face ABCD from where molecules rebound with an equal velocity.

24. Ans (1)

γ is vacuum; x is moving against no constrain So, no work done.

Container is insulated So, $\Delta Q = 0$

$\Delta Q = \Delta u + \Delta w$

$\Delta Q = \Delta w = 0 \Rightarrow \Delta u = 0 \Rightarrow \Delta T = 0$

Temperature remains same.

25. Ans (2)

$U = 5x(x - 4)$

$\Rightarrow F = -\frac{dU}{dx} = -10x + 20$

when $a = 0 \Rightarrow x = 2m$ velocity is maximum

$F = -10x + 20 \longrightarrow$ SHM eq^m

$T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{0.1}{10}} = \pi/5$ sec

26. Ans (1)

$f = \frac{n}{2L}\sqrt{\frac{250}{\mu}} = \frac{n+1}{2L}\sqrt{\frac{160}{\mu}}$

$5n = 4n + 4$

$n = 4$

$f = \frac{4}{2 \times 2}\sqrt{\frac{250}{10^{-3}}} = 500\,Hz$

$= 5 \times 10^2$ Hg $\Rightarrow f = 5$

Minimum no. of loop = 1

$\frac{4}{2L}\sqrt{\frac{250}{u}} = \frac{1}{2L}\sqrt{\frac{Mg}{u}}$

$20 = \sqrt{M}$

$M = 400$ kg

$= 4 \times 10^2$ kg $\Rightarrow n = 4$

So $n \times f = 4 \times 5 = 20$

27. Ans (2)

Horizontal range will be maximum when

$h = \frac{H}{2} = \frac{90}{2}$

= 45 cm i.e. hole 3.

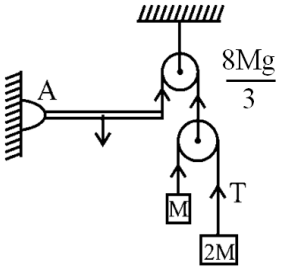
28. Ans (2)

When a block is placed in an open pan full of water:

If $\rho \leqslant \rho_w$**:** The block floats or stays submerged by displacing its own weight of water. Since the pan is open and full, the displaced water overflows. The total weight on that side of the pan remains unchanged, and equilibrium is maintained.

If $\rho > \rho_w$**:** The block sinks to the bottom. It displaces its volume of water, which overflows. However, because the block is denser than water, the weight of the block is greater than the weight of the displaced water. This creates an extra downward force on one side, destroying the equilibrium. Therefore, equilibrium is only maintained if the block does not add more weight than the water it displaces.

29. Ans (3)



$T = \frac{2M_1M_2g}{M_1 + M_2} = \frac{2(M)(2M)g}{3M}$

$T = \frac{4Mg}{3}$

Balancing torque

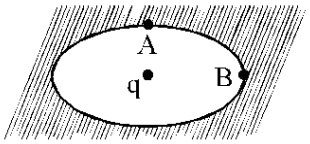
$M'g \times \frac{L}{2} = \frac{8Mg}{3} \times L$

$\frac{M'}{2} = \frac{8M}{3}$

$M' = \frac{16M}{3}$

30. Ans (3)

A and B points have different charge density due to different curvature.

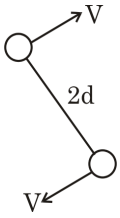


So $E_A \neq E_B$

$V_A = V_B$ because both are on same conductor.

Total flux = 0 because field inside bulk of conductor is zero.

31. Ans (3)



By energy conservation

$2 \times \frac{1}{2}mv_0^2 = 2 \times \frac{1}{2}mv^2 + \frac{kq^2}{2d}$

By angular momentum conservation

$2mv_0 \frac{d}{2} = 2mv\frac{2d}{2}$

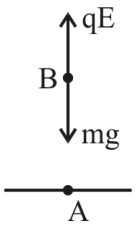
$\Rightarrow v = \frac{v_0}{2}$

$v_0 = \sqrt{\frac{q^2}{6\pi\varepsilon_0 md}}$

$\Delta KE = \frac{Kq^2}{2d} = \frac{q^2}{8\pi\varepsilon_0 d}$

32. Ans (1)

If B is pushed down, upwards electrostatic repulsion on it increases and hence it return back & vice versa



33. Ans (2)

Number for yellow is 4, Number of violet is 7

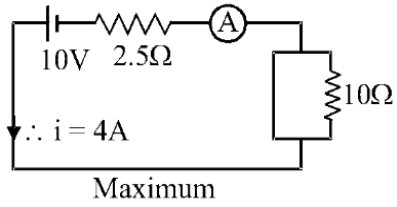
Brown colour gives multiplier 10^1,

Gold gives a tolerance of ± 5%

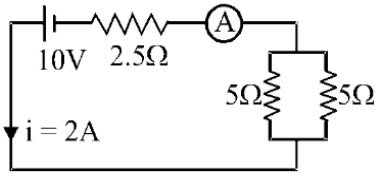
So resistance of resistor is $47 \times 10^1\ \Omega \pm 5\%$

or $470 \pm 5\%\ \Omega$.

34. Ans (4)



Maximum



Minimum

35. Ans (2)

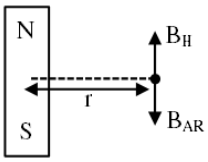
$a \rightarrow \frac{v}{4R}$ $\quad$ $a \rightarrow \frac{v}{4}$

$b \rightarrow \frac{v}{2R}$ $\quad$ $a \rightarrow \frac{v}{2}$

$c \rightarrow \frac{v}{3R}$ $\quad$ $C \longrightarrow 0$

$d \rightarrow \frac{v}{R}$ $\quad$ $D \longrightarrow 0$

36. Ans (1)



$$B_H = B_{AR} = \frac{KM}{r^3}$$

$$r = \left(\frac{10^{-7} \times 1.44}{18 \times 10^{-6}}\right)^{\frac{1}{3}} m = 0.2\ m = 20\ cm$$

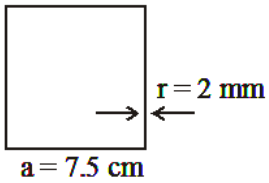
37. Ans (4)

$d\vec{F} = i\left(d\vec{\ell} \times \vec{B}\right)$

at all point of circumference

here $d\ell \parallel B$, hence $F = 0$

38. Ans (1)



$\ell = 4a$

$$V_i = \frac{d(Ba^2)}{dt} = a^2\frac{dB}{dt}$$

$$i = \frac{V_i}{R} = \frac{a^2 dB/dt}{\frac{\rho(\ell)}{\pi r^2}}$$

39. Ans (3)

$$X_C = \frac{2}{5}X_L$$

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

40. Ans (2)

$\vec{E} \times \vec{B} = \hat{K}$

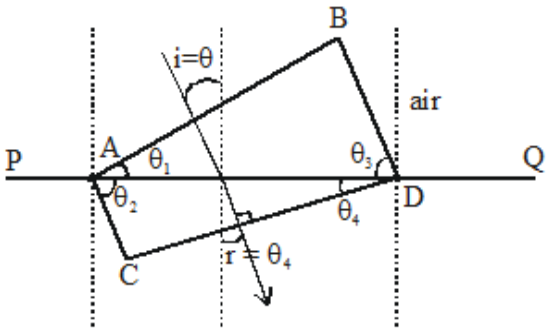
$\vec{E}.\vec{B} = 0$

41. Ans (3)

The shown figure represents (C) By Snell's law :

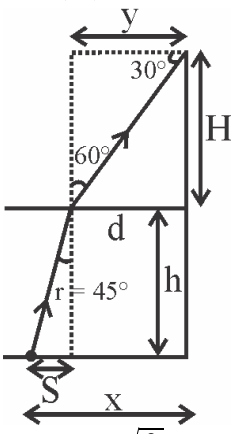
$$\frac{\sin i}{\sin r} = \frac{\mu_2}{\mu_1}$$

for $i = \theta_1$, $r = \theta_4$ and $\mu_1 = 1$



$$\mu_2 = \frac{\sin\theta_1}{\sin\theta_4}$$

42. Ans (4)



$\sin 60° = \sqrt{\frac{3}{2}}\sin r \quad r = 45°$

$S = h = 1m$

$y = H \tan 60° = 3m$

$\therefore x = S + y = 4m$

43. Ans (1)

$\text{Slope} = \tan\theta = \frac{m}{v}$

$m = \frac{f}{f+u} = \frac{f-v}{f}$

Focal length of the lens inverse of slop of line.

$m = \frac{f-v}{f}$

$m = 0$

$f = v$

Intercept on x-axis equal to focal length

$v = 0$

$m = \frac{f}{f} = 1$

Intercept on y-axis is equal to unity

Magnitude of slope of the line, Inverse of focal length

Final Answer: A(P); B(R); C(Q); D(S)

44. Ans (2)

$\Delta\phi = \frac{2\pi}{\lambda} d\sin\theta$

$= \frac{2\pi}{\lambda} d\frac{y}{D} = \frac{2\pi}{\lambda}\frac{d^2}{2D} = \frac{\pi}{D\lambda}d^2$

$\Delta\phi = \frac{\pi}{2}$

$I = I_0 + I_0 + 2\sqrt{I_0 I_0}\cos{}^{\pi}/_{2}$

$= 2I_0$

$\frac{\text{Intensity in front of slit}}{\text{Minimum Intensity on screen}} = \frac{2I_0}{4I_0} = \frac{1}{2}$

45. Ans (3)

Wave frent are $\perp$ to rays

46. Ans (1)

Question Explanation: Find how many pairs show the first species having a smaller ionization energy than the second.

Concept: IE decreases with increasing size.

Solution: Ionization Energy (IE) generally depends on the effective nuclear charge (Z_{eff}) and atomic size (r), following the relation $IE \propto \frac{Z_{eff}}{r}$.

• (v) Li and Li^+: The neutral atom Li has a lower IE than its cation Li^+ because the cation has a smaller size and a higher effective nuclear charge holding the remaining electrons more tightly.

• (vii) Ba and Sr : Both are in Group 2. Since Ba is below Sr, it has a larger atomic radius; therefore, its outermost electron is further from the nucleus and easier to remove.

For the other pairs: N > O (half-filled stability), Br > K (non-metal vs metal), Be > B (fully filled s -subshell), $I' > I^-$(neutral vs anion), and O > S (O is smaller).

47. Ans (4)

Question Explanation: Counting the number of double bonds and hydroxyl bonds in sulphur oxoacids.

Concept: Structural analysis of $H_2S_2O_8$and $H_2S_2O_7$

Solution:

- $H_2S_2O_8$:2 S atoms × 2 ($S = O$) = 4 ; 2 S atoms × 1 ($S - OH$) = 2.
- $H_2S_2O_7$: 2 S atoms × 2 ($S = O$) = 4 ; 2 S atoms × 1($S - OH$) = 2.

Final Answer: (4 and 2) and (4 and 2)

48. Ans (1)

ClH_3, FCl_3, XeF_3^-, HFO_2, $SiCl_6^{2-}$

49. Ans (4)

Bond order of CO^+ is > 3.

50. Ans (1)

$[Pt(NH_3)_4Cl_2][PtCl_4] \longrightarrow \underbrace{[Pt(NH_3)_4Cl_2]^{2+} + [PtCl_4]^{2-}}_{\text{2 complex ion}}$

$[Pt(NH_3)_4Cl_2][PtCl_4]$ — show GI and Co-ordination isomerism.

51. Ans (1)

Theory based.

52. Ans (1)

Explain Question :

Not explain by crystal field theory

Concept :

Base on CFT

Solution:

Crystal field theory introduce spectrochemical series based upon the experimental values of Δ but can't explain it's order. While other three points are explained by CFT. Specially when the CFSE increases thermodynamic stability of the complex increases.

Final answer : Option (1)

53. Ans (4)

O_2^- ⟶ Superoxide ligand ion

(Predicted on the basis of given t_{2g}^6 configuration)

54. Ans (3)

Question Explanation: Identifying the incorrect statement about manganate and permanganate ions.

Concept: Structure and Properties of Oxoanions.

Solution:

Option (1) In manganate and permanganate ions, the π-bonding takes place by overlap of p-orbitals of oxygen and d-orbitals of manganese, It's True

(2) MnO_4^{-2} ⇒ green

MnO_4^- ⇒ purple/violet

(3) Manganate contains 1 unpaired electron hence it is paramagnetic

where as permanganetic contains no unpaired electrons hence it is diamagnetic.

(4) Both have d^3s hybridisation hence both have tetrahedral geometry.

Final Answer: The correct option is (3)

55. Ans (1)

Question Explanation: Evaluating statements about the properties of transition metal oxides.

Concept: Properties of Metal Oxides.

Solution:

(1) statement is correct. These statements are factual.

(2) FeI_3 not exist.

(3) Along the period, the electronegativity of metals increases and therefore the difference of electronegativities between metal and oxygen decreases. Hence, the acidic character increases, so the correct order is TiO < VO < CrO < MnO.

(4) V_2O_5 reacts with alkalies as well as acids to give VO_4^{3-} and VO_2^+ respectively.

Final Answer: The correct options are (1)

56. Ans (3)

Question Explanation: $Cu(BO_2)_2$ (Blue) is reduced to $CuBO_2$(Colourless/Red opaque) in a luminous flame.

Concept: Borax Bead Test (Cu^{2+}).

Solution:

- **Statement I**
  (Luminous flame/Reducing flame): $Cu(BO_2)_2$ (Blue) is reduced to $CuBO_2$ or metallic Cu (Red opaque).

The colour change is from blue to red (opaque) or colourless. Stating the colour is green is **false**.

- **Statement II:** $Cu(BO_2)_2$ is CuII metaborate (Blue), not CuI metaborate. **False**.

Final Answer: Option

(3) Both Statement I and Statement II are false

Statement- I :- In this case, the expected colour change is not to green but rather to colourless or a different fue depending on the oxidation state of copper

Statement II:- The bead does not twin green, instead it may twin clourless to red due to the reduction of cuporic ions to cuprous ion metalic copper.

57. Ans (3)

Question Explanation: Identify which reaction does not produce a homonuclear diatomic gas.

Concept: Oxidation of halides and acidic strength.

Solution: In the presence of a strong oxidizing agent like MnO_2, halides are oxidized to their respective halogens.

However, NaCl reacting only with conc. H_2SO_4 (without an oxidizing agent) undergoes a double displacement reaction to produce HCl gas, which is a heteronuclear diatomic gas.

1. $2KI + MnO_2 + 2H_2SO_4 \xrightarrow{\Delta} MnSO_4 + K_2SO_4 + 2H_2O + I_2 \uparrow$

2. $2NaCl + MnO_2 + 2H_2SO_4 \xrightarrow{\Delta} MnSO_4 + Na_2SO_4 + 2H_2O + Cl_2 \uparrow$

3. $NaCl + H_2SO_4 \xrightarrow{\Delta} NaHSO_4 + HCl \uparrow$ (Homonuclear gas is not liberated)

4. $2NaBr + 2H_2SO_4 \xrightarrow{\Delta} Na_2SO_4 + SO_2 + 2H_2O + Br_2 \uparrow$ (Conc. H_2SO_4 is strong enough to oxidize Br^-)

Final Answer: Option (3)

58. Ans (3)

Pb^{2+} and Hg_2^{2+} are basic radicals of group (I)

59. Ans (1)

(A) Down the group; radius increases

(B) EN does not decrease gradually from C to Pb

(C) Correct

(D) Correct

(E) Range of oxidation state of carbon ; −4 to +4

60. Ans (4)

In trivalent state most of the compounds being covalent are hydrolysed in water. Trichlorides on hydrolysis in water form tetrahedral $[M(OH)_4]^-$ species, the hybridisation state of element M is sp^3.

In case of aluminium, acidified aqueous solution forms octahedral $[Al(H_2O)_6]^{3+}$ ion.

61. Ans (3)

$C_{(s)} + 2H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow CH_3OH_{(\ell)}; \Delta H^\circ = \Delta H_f^\circ$

$\Delta H^\circ = \Sigma(\Delta H^\circ_C)_R - \Sigma(\Delta H^\circ_C)_P$

or $\Delta H^\circ = [-393 + 2 \times (-286)] - (-726)$

$= -965 + 726 = -239$ kJ mol^{-1}

62. Ans (3)

Isothermal

$\Delta H = nC_{p,m}\,\Delta T$

So $\boxed{\Delta H = 0}$

63. Ans (4)

Question Explanation:

To find out the number of electron transferred per mole in given reaction.

Concept:

Determination of oxidation state.

Solution:

(A) $\overset{+3+5}{Fe}(NO_3)_3 \rightarrow \overset{+2+3}{Fe}(NO_2)_2$

Total e^- gained = 1 + 3 × 2 = 7 (R)

(B) $\overset{+1-2}{CuS} \rightarrow \overset{+2}{Cu^{2+}} + \overset{+4}{SO_2}$

Total e^- lost = 1 × 2 + 6 × 1 = 8(S)

(C) $\overset{-2}{N_2}H_4 \rightarrow \overset{0}{N_2}$

Total e^- lost = 2 × 2 = 4 (Q)

(D) $KI^{-1} \rightarrow \overset{+5}{K}IO_3$

Total e^- lost = 6 × 1 = 6 (P)

Final answer: Option(4)

64. Ans (4)

Radial node = $n - \ell - 1$

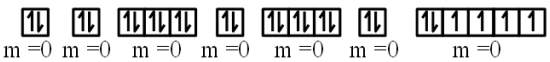
For 2s = 2 − 0 − 1 = 1

In (4) 2 Radial

node in 2s which is not posible

65. Ans (1)

Fe = $1s^2$ $2s^2$ $2p^6$ $3s^2$ $3p^6$ $4s^2$ $3d^6$



m = 0 m = 0 m = 0 m = 0 m = 0 m = 0 m = 0

$n + \ell = 5 \Rightarrow 3d \Rightarrow 6e$

66. Ans (2)

$\Delta G = -nF\,E_{cell}$

$= -3 \times 96\,500 \times 0.560$

$= -1.62 \times 10^5$ joule

For $\frac{1}{2}$mol Cl_2 n = 3

67. Ans (2)

$E_{cell} = 0.059\ (pH_A - pH_C)$

$\therefore\ pH_A < pH_C$

$\therefore\ E_{cell} = -\ ve$

$\Delta G = nFE_{cell}$

$\Delta G > 0$

so non-spontaneous

68. Ans (2)

$\Delta T_f = i\ k_f.m$

$i = 1 + (n - 1)\ \alpha$

$= 1 + (2 - 1)\ 0.2 = 1.2$

$\Rightarrow \Delta T_f = 1.2 \times 1.86 \times 0.2$

$= 0.4464$

$\approx 0.45°C$

So $T_f = -\ 0.45°C$

69. Ans (2)

$i = 1 - \alpha\ (n - 1)$

$1 + 0.75\ (5 - 1) = 4$

$\pi = iCRT$

$= \frac{1}{10} \times 0.08 \times 500 \times 4$

$\simeq 16$

70. Ans (2)

Explain the Question : We must calculate the rate constant k for the first-order decomposition

$2A(g) \longrightarrow B(g) + C(s)$ using total-pressure data.

Concept : This question is based on first-order kinetics

Solution :

Reaction:	$2A(g) \longrightarrow$	$B(g)$ +	$C(s)$
Initial pressure	P_0	0	0
Pressure at t =10 min	$P_0 - x$	x/2	0

Total pressure at time t:

$P_t = (P_0 - x) + x/2 = P_0 - x/2$

Given:

At completion:

$P_t = 200 \longrightarrow P_0/2 = 200 \longrightarrow P_0 = 400$

At t = 10 min:

$P_t = 300 \longrightarrow 400 - x/2 = 300 \longrightarrow x/2 = 100 \longrightarrow x = 200$

Pressure of A at 10 min

$= P_0 - x = 400 - 200 = 200$

First-order formula :

$2.3 \log(P_0 / (P_0 - x)) = k\ t$

$2.3 \log(400 / 200) = k \times 10$

$k = 0.693 / 10 = 0.0693\ min^{-1}$

Final Answer : (2)

71. Ans (4)

Explain Question:

Analyzing the shift in the Maxwell-Boltzmann distribution of kinetic energies when temperature increases.

Concept:

Maxwell-Boltzmann distribution and temperature.

Solution:

The Maxwell-Boltzmann distribution curve shows how many molecules in a sample prowess a certain amount of kinetic energy. 1 when the temperature (T) is increased, the following changes occur:

(1) Shift to the right: the most probable kinetic energy increases because the molecules move faster on average. The peak of the graph moves toward higher energy values.

(2)Flattening on the peak: since the total number of molecules (the area under the curve) must remain constant, the peak must become lower as the curve spreads out over a wider range of energies.

(3)Area I vs. Area II:

- Area I (Low-energy region): As the curve flattens and shift right, the proportion of molecules with lower kinetic energy decreases.
- Area II (High-energy region) The '' tail'' of the curve on the right side becomes higher, representing an increase in the fraction of molecules with high kinetic energy (specifically those exceeding the activation energy).

Final Answer: option (4)

72. Ans (3)

Conditions of collision theory.

73. Ans (1)

Question Explanation:
This question matches kinetic theory terms with their correct descriptions related to the Arrhenius equation.

Concept:
This question is based on collision theory

Solution:

- A ⟶ Number of collisions per second per unit volume = collision frequency (q)
- B ⟶ Fraction of molecules with energy ≥ E_a = $e^{-Ea/RT}$ (r)
- C ⟶ Arrhenius coefficient = frequency factor (s)
- D ⟶ Collision with sufficient K.E. and proper orientation = effective collision (p)

Therefore: A – (q), B – (r), C – (s), D – (p)

Final Answer: Option 1

74. Ans (1)

Question Explanation :
The question asks to calculate Kp for the gaseous equilibrium.

Concept :
Gaseous equilibrium and Kp in terms of degree of Dissociation.

Solution :

$$\underset{1}{N_2O_3(g)} \rightleftharpoons \underset{0}{NO(g)} + \underset{0}{NO_2(g)}$$

$$\underset{1-\alpha}{} \qquad \underset{\alpha}{} \qquad \underset{\alpha}{}$$

Total moles = $1 + \alpha$

Partial pressures:

$$P_{(NO)} = \frac{\alpha}{(1+\alpha)} \cdot P$$

$$P_{(NO_2)} = \frac{\alpha}{(1+\alpha)} \cdot P$$

$$P_{(N_2O_3)} = \frac{(1-\alpha)}{(1+\alpha)} \cdot P$$

Kp expression:

$$Kp = \frac{\left(P_{(NO)} \times P_{(NO_2)}\right)}{P_{(N_2O_3)}}$$

Substitute values:

$$Kp = \frac{\left[\left(\frac{\alpha}{(1+\alpha)} \cdot P\right)^2\right]}{\left[\frac{(1-\alpha)}{(1+\alpha)} \cdot P\right]}$$

$$Kp = \frac{\alpha^2 P}{(1-\alpha^2)}$$

Given $\alpha = \sqrt{0.5} \Rightarrow \alpha^2 = 0.5$

$$Kp = \frac{(0.5\,P)}{(1-0.5)}$$

$Kp = P$

Final Answer : Option (1)

75. Ans (2)

Question Explanation:

Decide if $BaSO_4$ precipitates by comparing ionic product after mixing each pair to $K_{sp} = 10 \times 10^{-10}$. Show how final concentration are obtained.

Concept:

this question is based on dilute on mixing equal volumes and ionic product vs K_{sp}

Solution:

- Final concentration on mixing (100 mL + 100 mL) : Total volume = 200 mL ⇒ each solute's concentration becomes half ($C_{final} = ½ \times C_{initial}$).

Case 1 : $[Ba^{2+}] = 2 \times 10^{-4}$, $[SO_4^{-2}] = 3.0 \times 10^{-8}$

$\Rightarrow 6 \times 10^{-12} < K_{sp} \longrightarrow$ no ppt

Case 2 : $[Ba^{2+}] = 2 \times 10^{-3}$, $[SO_4^{-2}] = 3.0 \times 10^{-4}$

$\Rightarrow 6 \times 10^{-7} < K_{sp} \longrightarrow$ ppt forms.

Case 3 : $[Ba^{2+}] = 2 \times 10^{-4}$, $[SO_4^{-2}] = 3.0 \times 10^{-6}$

$\Rightarrow 6 \times 10^{-10} < 11 \times 10^{-10} \longrightarrow$ no ppt

Final Answer: Option (2)

76. Ans (3)

Question Explanation: Apply subtraction rule of significant figures (least decimal places).

Concept : significant figures

In addition/subtraction, result is reported to the least number of decimal places.

Solution:

41.6325 – 41.6120 = **0.0205**

Now apply rule:

41.612 has **3 decimal places** ⟶ Result must be rounded to **3 decimal places**

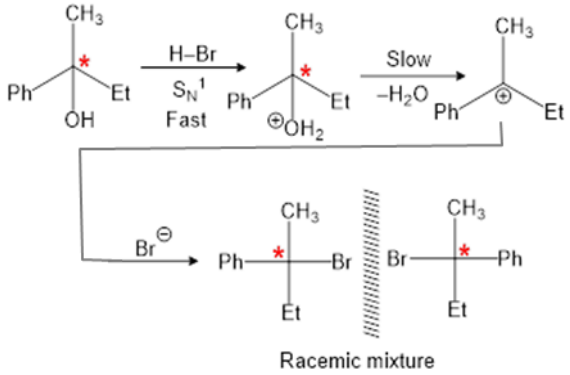
So, 0.0205 ⟶ **0.020**

Significant figures in 0.020 = **2**

Final Answer: Option (3)

77. **Ans (3)**

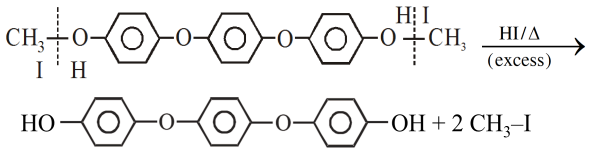
Solution: The reaction of a tertiary alcohol with a strong acid like HBr proceeds via the SN1 mechanism. The OH group is protonated, and water leaves, forming a stable 3° carbocation intermediate. The 3° carbocation is planar. The nucleophile (Br−) can attack from either side with equal probability. This leads to a racemic mixture (racemization).



Final answer: option (3)

78. **Ans (2)**

Solution:



2 moles of HI consumed in the above reaction.

Final answer: option (2)

79. **Ans (2)**

Solution:

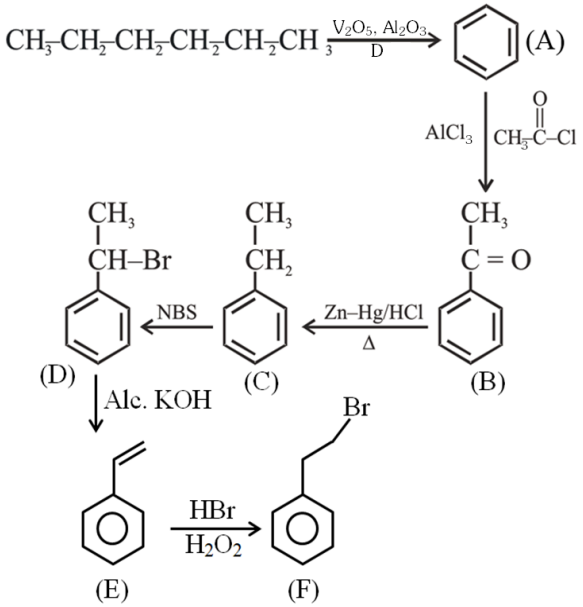
Pair (Column-I)	Reagent (Column-II)	Test Principle
(i) Terminal Alkyne vs Alkene	(e) $AgNO_3$ + NH_4OH	Acidic H on alkyne forms White Ppt; Alkene does not react.
(ii) 2° vs 1° Alcohol	(a) HCl + $ZnCl_2$	Lucas Test: 2° forms turbidity (5 min); 1° does not at RT.
(iii) Phenol vs Carboxylic Acid	(d) $NaHCO_3$	Strong acids evolve CO_2 gas; Phenols are too weak to react.
(iv) 1° vs 2° Amine	(c) $CHCl_3$ + Alc. KOH	Carbylamine Test : Only 1° amines produce foul smelling isocyanide.

Correct Match: (i)-e, (ii)-a, (iii)-d, (iv)-c

Final Answer: Option (2)

80. **Ans (2)**

Solution :



Final answer : 2

81. **Ans (3)**

All NCERT Rx.

82. **Ans (3)**

(a) Rosenmund reduction uses H_2/Pd-$BaSO_4$.

(b) HVZ reaction Cl_2/Red P, H_2O

(c) Hoffmann bromide degradation Br_2/NaOH

(d) Clemmensen reduction uses Zn(Hg)/HCl.

Correct Match is: (a)−(ii), (b)−(iv), (c)−(i), (d)−(iii)

Final Answer: Option (3)

83. **Ans (4)**

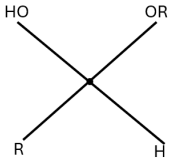
NCERT-XIIth, Biomolecules

84. **Ans (4)**

Haloform Rx (XIIth NCERT)

85. Ans (4)

Solution:

⇒ An any with a hemiacetal  are is

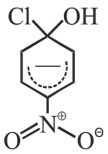
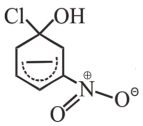
equilibrium with a ring opened form containing an aldehyde and will react as a reducing agent towards certain (oxidizing) metal salts such as Cu^{2+} (Benedict's test, Fehling's solution) and Ag^{+} (tollen's test) are there fore called as reducing sugar.

⇒ Since all sugar are with hemiacetal group, hence all are reducing sugars.

Final Answer : Option: (4)

86. Ans (2)

Solution :-

Intermediates	 Intermediate-A	 Intermediate-B
Effects	**–I and –M effect of $-NO_2$**	**Only –I effect of $-NO_2$**

Stability of anion = A > B

Final Answer :- 2

87. Ans (1)

Solution : Option 1 (ortho > para > meta) — Incorrect: Actual order is para > ortho > meta due to stronger −M and −I effects at para stabilizing phenoxide. Ortho H-bonding reduces resonance, meta has weakest stabilization.

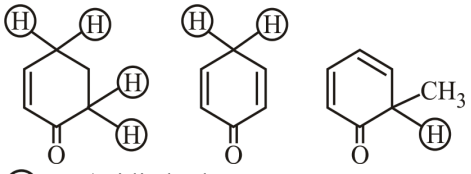
Option 2 ($F_3C - COOH > F_2CH - COOH > F - CH_2 - COOH$) — Correct: More fluorines increase −I effect, raising acidity.

Option 3 ($H - COOH > Ph - COOH > CH_3COOH$) — Correct : Formic acid most acidic; benzoic acid next; acetic acid least due to +I effect.

Option 4 ($Ph - OH > C_2H_5OH > CH \equiv CH$) — Correct : Phenol > ethanol > terminal alkyne acidity order.

Final answer : 1

88. Ans (4)



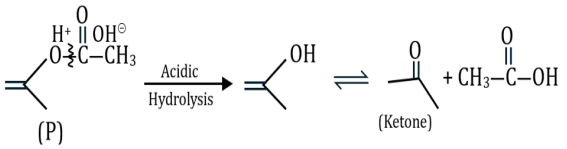
Ⓗ ⇒ Acidic hydrogen

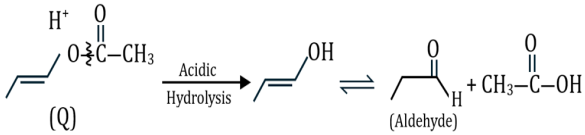
89. Ans (3)

II, IV are G.I. other not show G.I.

90. Ans (4)

Solution :





∵ Aldehyde & ketone can differentiated by Fehling solution because aldehyde give Fehling test while ketone doesn't give Fehling test.

Final Answer : Option (4)

91. Ans (4)

NCERT-XI, Pg # 4

92. Ans (4)

NCERT XI Page No. # 11

93. Ans (1)

NCERT-XI, Page No. # 90, 91

94. Ans (3)

NCERT-XI, Page No. # 27, 28

95. Ans (2)

NCERT-XI, Page No. # 28

96. Ans (3)

NCERT XI Page No. # 27,28,29,32,33

97. Ans (4)

Module - 2

98. Ans (3)

Module - 2

99. Ans (3)

NCERT Pg. No.# 61,63,64,65

100. Ans (1)
NCERT Page No. 73

101. Ans (2)
NCERT Page No. 76

102. Ans (3)
NCERT XI, Page No. # 76, 77

103. Ans (4)
NCERT Pg.#5.

104. Ans (1)
NCERT-XII, Pg. No. # 15

105. Ans (1)
NCERT XII 9, 10, 11

106. Ans (3)
NCERT Pg. No. # 95

107. Ans (4)
NCERT-XI, Pg # 101

108. Ans (4)
NCERT XI Pag No.# 91, 94, 96, 97

109. Ans (4)
NCERT-XI Pg. # 126

110. Ans (3)
NCERT Pg. # 121

111. Ans (2)
NCERT-XI Pg. # 123

112. Ans (4)
NCERT XI, Pg. # 123, 124, 127

113. Ans (4)
NCERT Pg. # 73

114. Ans (3)
NCERT, Pg # 74,75,76

115. Ans (1)
NCERT XII, Pg. # 78

116. Ans (1)
NCERT (XII) ; Page No. # 70,71

117. Ans (1)
NCERT-XII Pg # 84

118. Ans (3)
NCERT Pg.# 105

119. Ans (4)
NCERT-XII, Pg. No. # 100

120. Ans (1)
NCERT-XII, Pg. No. # 92

121. Ans (4)
NCERT-XII Pg. # 98, 99, 96, 90

122. Ans (1)
NCERT, Pg. # 168

123. Ans (3)
NCERT, Pg. # 169

124. Ans (3)
NCERT Pg.# 170

125. Ans (2)
NCERT Pg. # 166-167, 179, 181, 182

126. Ans (3)
NCERT (XII) ; Page No. # 180

127. Ans (3)
NCERT-XII, Pg. # 184, 185 (E), 202, 203 (H)

128. Ans (3)
NCERT (XII) Page No. # 172

129. Ans (2)
NCERT Pg.# 151-152

130. Ans (2)
NCERT Pg.# 154

131. Ans (1)
NCERT (XII) ; Page No. # 198

132. Ans (2)
NCERT, Pg.# 198, 200, 201

133. Ans (1)
NCERT Pg. No. # 196

134. Ans (3)
NCERT Pg.# 208, 210, 211

135. Ans (1)
NCERT Pg.# 211

136. Ans (3)
New, NCERT, Pg. # Module

137. Ans (4)
NCERT Pg. # 224

138. Ans (2)
NCERT, Pg. # 220

139. Ans (4)
NCERT-XIIth, Pg. # 220

140. Ans (1)
Case 1 : Mother will be homozygous diseased.
Case 2 : Disease is autosomal, so son and daughter can be affected.

141. Ans (2)
NCERT XII Pg.no. 143

142. Ans (4)
NCERT-Pg # 136, 137

143. Ans (3)
NCERT Page No. 130-132

144. Ans (2)
NCERT (XII) Pg. # 111, 112, 119, 121

145. Ans (2)
NCERT Page No. 120
In Hardy-weinberg's principle : -
frequency of heterozygous is 2pq.
(A) = p = 0.19
(a) = q = 0.81
$\Rightarrow 2 \times 0.19 \times 0.81$
$\Rightarrow 0.3.78$
% of heterozygous individual = $0.3078 \times 100 = 30.78$
$\Rightarrow \sim 31\%$

146. Ans (2)
NCERT XII, Pg # 48

147. Ans (3)
Ref. NCERT XII, pg.# 31

148. Ans (4)
NCERT Pg. # 34

149. Ans (3)
NCERT Pg. # 33

150. Ans (3)
Module No. 6 Page. No.63

151. Ans (1)
NCERT Pg # 199

152. Ans (1)
NCERT Page No. # 275, 276

153. Ans (4)
NCERT XI, Pg. # 187

154. Ans (1)
NCERT-XI, Pg. No. # 241

155. Ans (4)
XI NCERT Pg. No :- 331, 332

156. Ans (4)
NCERT Pg # 243 (19.2.4)

157. Ans (4)
NCERT XII Pg. # 317

158. Ans (3)
NCERT XI Pg.#236

159. Ans (2)
XIth NCERT Page No. – 298

160. Ans (3)
Module-5 Page No.#66

161. Ans (4)
NCERT-XI, Page No. 308, Fig. 20.5

162. Ans (2)
NCERT-XI,, Pg. # 104, 105,108, 109

163. Ans (3)
NCERT-XI, Pg. # 106,108,109,110

164. Ans (1)
NCERT-XI, Pg. # 105

165. Ans (3)
NCERT-XI, Pg. #112, 113, 115

166. Ans (1)
NCERT-XI, Pg. # 146, 147

167. Ans (1)
NCERT-XI, Pg. # 143, 144

168. Ans (3)

NCERT-XI, Pg. # 163

169. Ans (3)

NCERT-XI, Pg. # 159, 160

Complete oxidation of acetyl CoA is krebs cycle.

Two krebs cycle ⇒

6 $NADH_2$ = 18 ATP

2 $FADH_2$ = 4 ATP

2 GTP = 2 ATP

24 ATP

170. Ans (2)

NCERT-XI, Pg. # 172

171. Ans (1)

NCERT-XI, Pg # 175, 176, 177, 178

172. Ans (3)

NCERT-XI, Pg.# 174 (Fig. 13.10)

The figure 'C' is with opaque cap and figure 'd' is with transparent cap.

173. Ans (3)

Module

174. Ans (1)

Module

175. Ans (1)

NCERT-XI, Pg. # 112,113

176. Ans (1)

NCERT-XI Pg. # 119

177. Ans (1)

NCERT Page No.# 56

178. Ans (2)

NCERT Pg.# 46-48

179. Ans (4)

Module No. # 4, Pg. No. # 15

180. Ans (4)

NCERT Pg. # 40-43